

Multi-Criteria in Selecting a Project

(Larson & Gray, 2018)

- The Custom Bike Company has set up a weighted scoring matrix for evaluation of potential projects. Below are five projects under consideration.
 - Using the scoring matrix in the following chart, which project would you rate highest? Lowest?

Highest: 5
Lowest: 2
 - If the weight for "Strong Sponsor" is changed from 2.0 to 5.0, will the project selection change? What are the three highest weighted project scores with this new weight?

$3 > 5 > 4 > 1 > 2$
 - Why is it important that the weights mirror critical strategic factors?

Project Screening Matrix							
Criteria Weight	Strong sponsor	Supports business strategy	Urgency	10% of sales from new products	Competition	Fill market gap	Weighted total
	2.0	5.0	4.0	3.0	1.0	3.0	
Project 1	9	5	2	0	2	5	68
Project 2	3	7	2	0	5	1	57
Project 3	6	8	2	3	6	8	99
Project 4	1	0	5	10	6	9	85
Project 5	3	10	10	1	8	0	107

StrongSponsor = 5
95
66
117
88
116

- Please help Mr. Quan to choose between four kinds of car (A,B,C,D) for his transportation.

Criteria	Kind of car				Weight (%)
	A	B	C	D	
Min Z1	30	40	50	60	20
Max Z2	20.000	10.000	15.000	30.000	25
Max Z3	0,5	0,8	1,5	2,0	10
Max Z4	2	5	3	10	30
Min Z5	3	5	5	9	15
CU=	0.34	0.33	0.35	0.25	

- Find the dominated alternative(s) if any. B, C
- Use the method of Collective Utility to rank and find the best alternative. $C > A > B > D$
- Use the method of Compromise Programming (shortest distance) to rank and find the best alternative.

- The sign $X \rightarrow Y$ indicates that the alternate X is more favourable than Y. Choose the group of nondominant alternatives (Kernel), given that Kernel: {C, B, F}

$C \rightarrow A$; $A \rightarrow B$; $C \rightarrow E$; $B \rightarrow D$; $D \rightarrow F$; $E \rightarrow F$; and $F \rightarrow G$